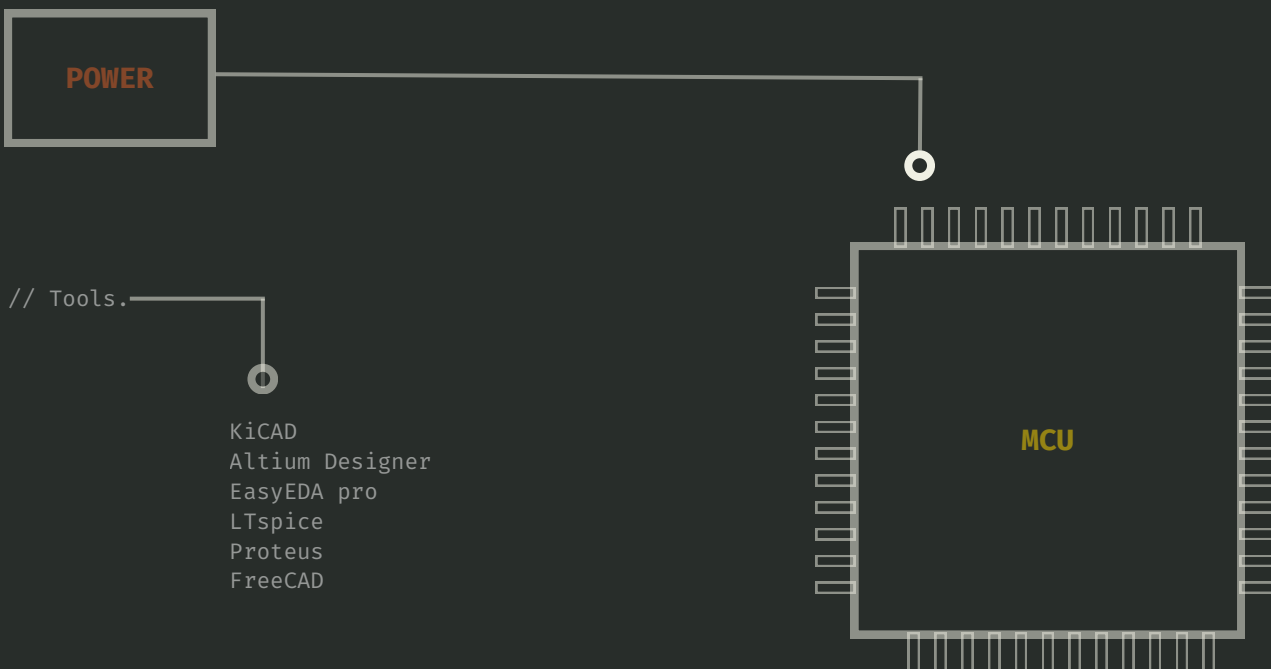
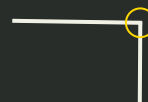


// Engineering Portfolio.



// Engineering Portfolio.

ATIF KHAN

EMBEDDED HARDWARE DESIGN ENGINEER

Schematics, PCB layout, and firmware bring-up, turning circuit ideas into reliable, manufacturable boards.

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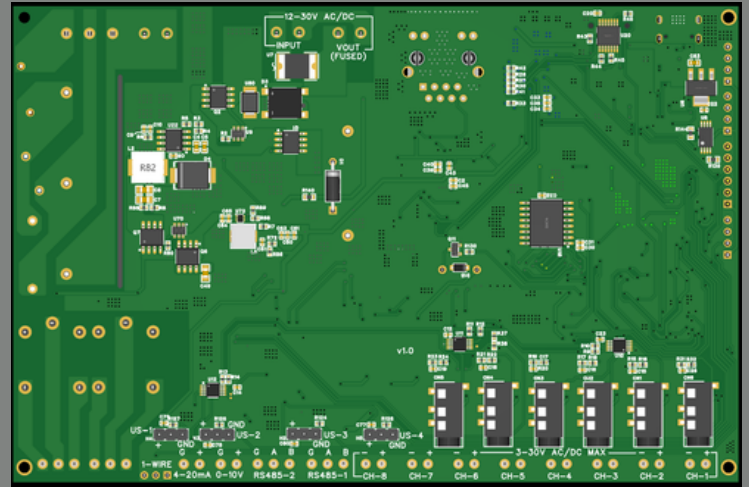
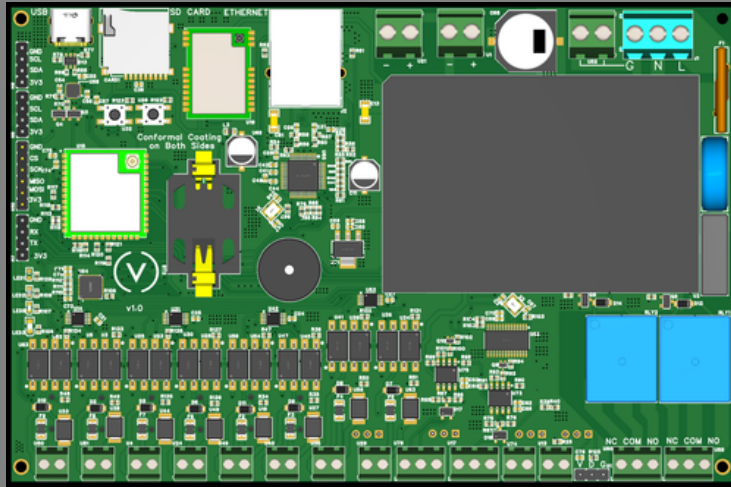
// INTRODUCTION.

Embedded Hardware Design Engineer with 4+ years of experience spanning schematic design, multilayer PCB layout, and embedded systems development for industrial, IoT, wearable, and consumer electronics products. Proven expertise in power electronics architecture DC-DC conversion, battery management systems (BMS), and power-path switching alongside RF front-end design and ATEX/IECEX-compliant hardware for hazardous-area applications. Proficient in Altium Designer and KiCad for high speed, impedance-controlled, multilayer board design, with hands-on experience in MCU bring-up, hardware-in-the-loop (HIL) validation, and EMI/EMC mitigation. Holds a Bachelor of Science in Engineering Sciences (Optoelectronics) from GIKI. Skilled at translating client and product requirements into reliable, manufacturable, production ready hardware from concept through certification.

TECHNICAL SKILLS

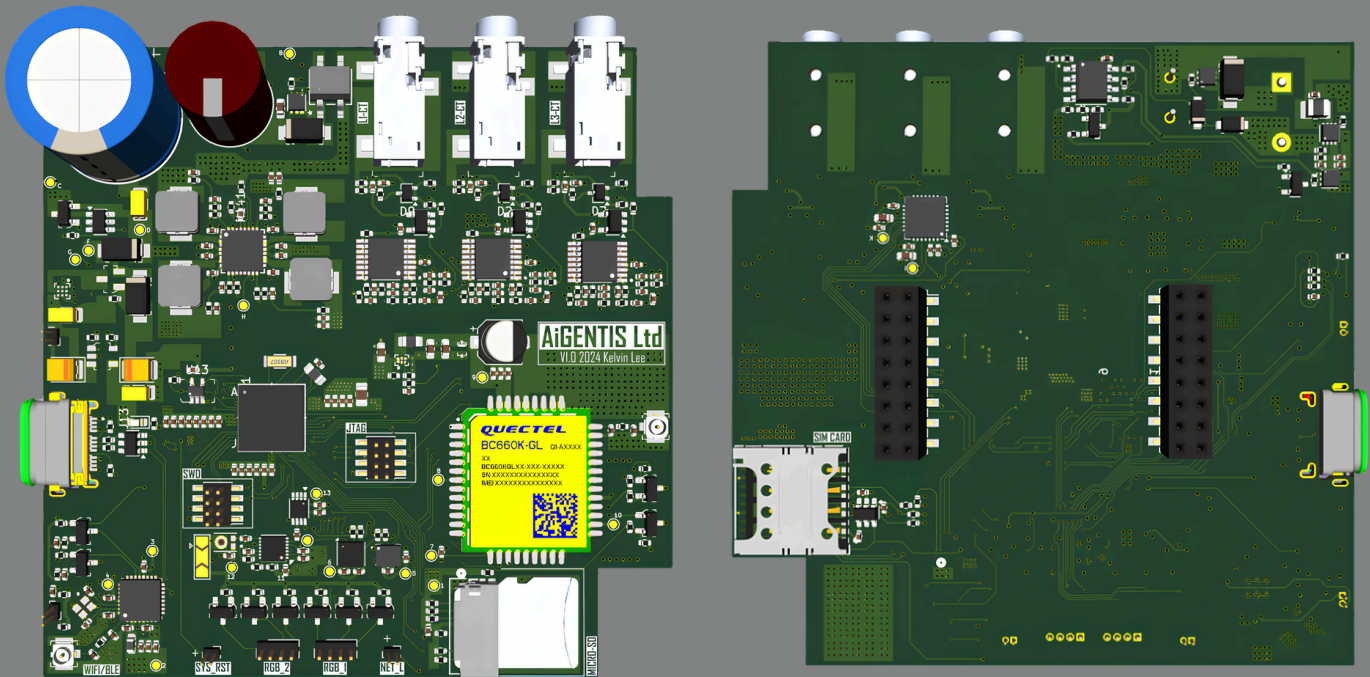
- **EDA / CAD Tools:** Altium Designer, KiCad 9.0, EasyEDA Pro, LTspice, Proteus, SolidWorks, FreeCAD
- **Microcontrollers & SoCs:** STM32, ESP32 / ESP32-S3 / ESP32-C6, Nordic nRF52840, AVR (ATtiny, ATmega), AT89S52 (8051), PIC, Renesas Synergy, MAX78000 edge-AI SoC
- **Communication Protocols & Interfaces:** SPI, I2C, UART, CAN, RS-232, RS-485, USB, Ethernet, JTAG/SWD, UPDI
- **Wireless / IoT:** Quectel & SIMCom LTE-M/NB-IoT modules, BLE 5.0, Wi-Fi 6, GNSS, NFC, LORA.
- **Programming & Simulation:** C, C++, Python, Embedded C, MATLAB, Simulink
- **Other:** Microsoft Office, WordPress, HTML/CSS, technical documentation, project management





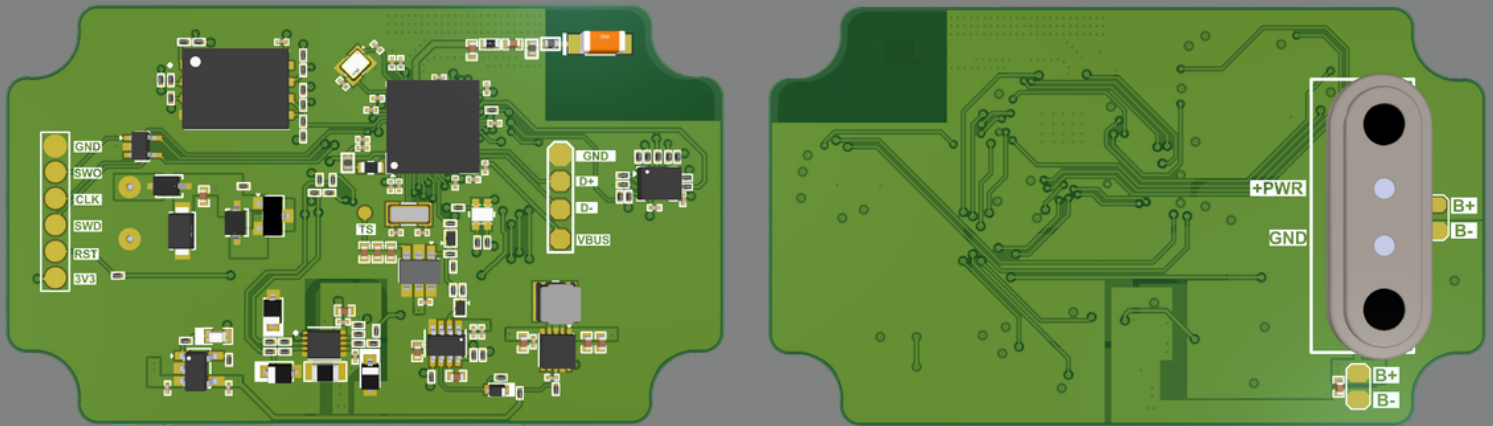
// Description:

- **Wide-range universal power input:** Designed dual-mode power entry accepting 120VAC or 12-30V AC/DC, with full-bridge rectification, TVS surge suppression (SMBJ33CA), and PPTC fuse protection ahead of regulation.
- **Automatic power-source arbitration:** Implemented dual LTC4412 ideal-diode controllers driving back-to-back IRF7416 MOSFETs for seamless, lossless switchover between two independent DC input sources without diode forward-drop losses.
- **Multi-rail regulated power tree:** Built a cascaded regulation architecture (TPS54360 high-current buck → TPS564257 secondary buck → AMS1117 LDOs) to generate clean +5V, +3V3, and +3V3_ESP domains from a wide unregulated input range.
- **ESP32-S3 core with USB/battery power muxing:** Integrated a TPS2113A ideal-diode power multiplexer to auto-select between USB-C bus power and system rail power for the ESP32-S3 MCU, with CH343P USB-to-UART bridge and USBLC6-2SC6 ESD protection on the USB-C data lines.
- **Precision analog acquisition front-end:** Deployed three ADS1115 16-bit I2C ADCs (unique addresses 0x48-0x4A) to support both 0-10V and 4-20mA industrial sensor inputs with per-channel signal conditioning.
- **Long-range wireless telemetry:** Integrated an E22-900M22S (SX1262-based) LoRa sub-GHz radio over SPI with matched RF front-end and a U.FL-to-SMA external antenna interface for extended-range communication.
- **Isolated 8-channel I/O bank:** Designed an optically isolated I/O expansion stage using PC817 optocouplers paired with TS5A23157 analog switches, giving each of 8 channels independently configurable sink/source behavior with TVS-protected field wiring.
- **I2C-based system expansion and status indication:** Used a TCA6424 24-bit I2C GPIO expander to manage status LEDs, an audible buzzer, and relay control lines, reducing MCU pin usage while centralizing system status feedback.
- **Wired Ethernet and RS-485 connectivity:** Added a W5500 SPI-to-Ethernet controller (magnetics-isolated RJ45) alongside a dual-channel SC16IS752 UART bridge driving two MAX3485 RS-485 transceivers, giving the platform redundant wired networking and industrial serial bus support.
- **High-density 2-layer layout with controlled differential routing:** Successfully routed a 13-sheet, multi-domain design (Ethernet, USB, RS-485, and RF differential pairs alongside mixed-voltage power planes) on a 2-layer PCB while maintaining differential pair length-matching and impedance discipline.



// Description:

- Designed a **6-layer**, hierarchical mixed-signal **PCB in KiCad 9.0** for an **edge-AI electrical/arc-fault detection sensor node**
- Integrated **MAX78000 (Arm Cortex-M4F + CNN hardware accelerator)** for real-time, on-device fault classification – no cloud dependency
- Built an **arc-fault analog front-end based on TI's TIDA-01385/TIDA-010231** reference design – CT sensing, OPA4322 active filtering, TLV3492 comparators, MAX11122 12-bit SAR ADC
- Implemented **CT-based energy harvesting with TPS61021A** boost converter and **supercapacitor storage** for self-powered field operation
- Designed automatic **power-path switching (LTC4412)** across USB, Li-Ion battery, and harvested energy sources
- Architected a **4-rail power system** (1.2V/1.8V/2.5V/3.3V) using the LTC3370 configurable quad buck converter
- Added full **battery management: MCP73831 Li-Ion charger** + BQ27427 **fuel gauge** for accurate **state-of-charge** tracking
- Enabled **dual-radio IoT connectivity: Quectel BC660K-GL (LTE-M/NB-IoT)** for WAN + **ESP32-C6 (Wi-Fi 6/BLE)** for local wireless
- Fused multi-modal sensor data – BMI270 IMU (vibration), BME690 (environmental), MEMS mic + MAX9814/ADS7042 (acoustic arc signatures)
- Included SPI microSD data logging, JTAG/SWD debug headers, and extensive test points for hardware validation on a dense 6-layer stack-up

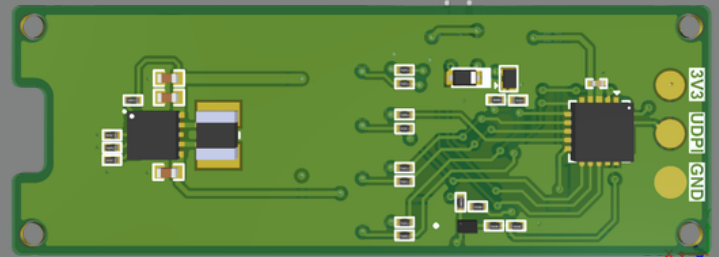
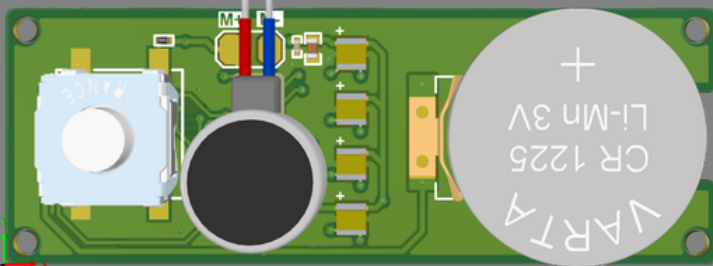


// Description:

- **Wireless MCU/SoC:** **Nordic nRF52840** (Arm Cortex-M4F + 2.4 GHz radio) – Bluetooth Low Energy (**BLE 5**) connectivity for real-time streaming to a phone/watch app, with **NFC tap-to-pair support**.
- **6-Axis Motion Sensing:** LSM6DSV16X IMU (accelerometer + gyroscope) for capturing stride, cadence, posture, and motion-intensity data directly from the chest – engineered for stable signal capture during high-movement activity.
- **Battery Management & Fuel Gauging:** MCP73831 Li-Po charger paired with the MAX17055 ModelGauge fuel gauge IC for precise state-of-charge and runtime estimation, critical for long-duration training sessions.
- **Efficient Power Architecture:** TPS63031 buck-boost converter for stable regulated output across the full battery discharge curve, plus a LTC2954 pushbutton power controller for clean system on/off and kill-switch functionality.
- **On-Board Data Logging:** NAND flash storage (ZSD01GLGEAG, SD-protocol interface) for buffering workout data locally when out of BLE range, enabling offline-capable training sessions.
- **USB-C Charging Interface:** Modern USB Type-C connector for charging and firmware programming, with **ESD protection** on the data lines.
- **Human-Machine Interface:** **Capacitive touch** buttons (no moving parts – ideal for a sweat/sealed wearable) and an RGB status LED for charge/connection/alert indication.
- **RF & Timing:** Custom-matched **2.4 GHz antenna** front-end and dual crystal oscillators (32.768 kHz RTC + 32 MHz radio reference) for accurate timekeeping and reliable wireless link performance.
- **Debug & Manufacturing:** Dedicated SWD programming header and test points for production test and field debugging.
- The design was fully captured and routed in **Altium Designer**, with a compact **4-layer stack-up** optimized for a small, wearable footprint, RF performance, and battery-powered operation.

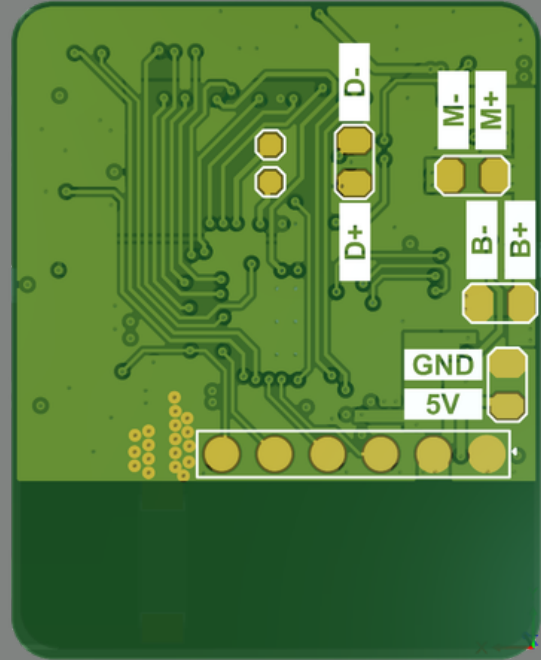
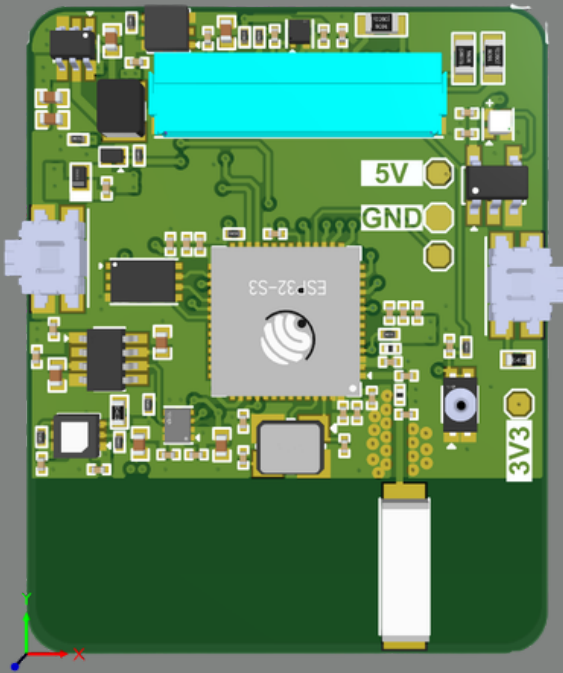


- Designed a **4-layer PCB in KiCad 9.0** for KIPFLO – a **smart, motor-actuated liquid dispensing**/flow-control device with integrated sensing and wireless control
- Built around the **ESP32-S3** (dual-core, Wi-Fi/BLE SoC) as the central wireless MCU, handling sensor fusion, motor control, and connectivity
- Implemented a **dual H-bridge motor driver (DRV8833)** in **parallel-bridge mode** with overcurrent, thermal shutdown, and shoot-through protection for precise flow-actuation control
- Integrated digital latching **Hall-effect sensors (DRV5013)** for contactless position/rotation sensing of the mechanical flow mechanism
- Designed a **capacitive liquid-sensing** front end using electrode-based conductivity **sensing for real-time fluid presence/level detection**
- Added a **dedicated spout detection circuit** for automatic state-awareness of the dispensing mechanism
- Integrated a **3-axis accelerometer (BMA400)** for orientation/tilt detection, motion-triggered automation, and fall/usage-safety logic
- Implemented a complete battery charging system using the BQ24074 linear charger with **Dynamic Power Path Management (DPPM), input OVP, and status LED indication**
- Designed a **buck-boost power supply (TPS63031)** for stable voltage regulation across the full Li-ion battery discharge range, with USB-C ESD protection (USBLC6-2SC6)
- Engineered a **POGO-pin charging dock interface** with **automatic power-path switching – system shuts down** on dock connection and powers on only when undocked, for safe wireless charging behavior



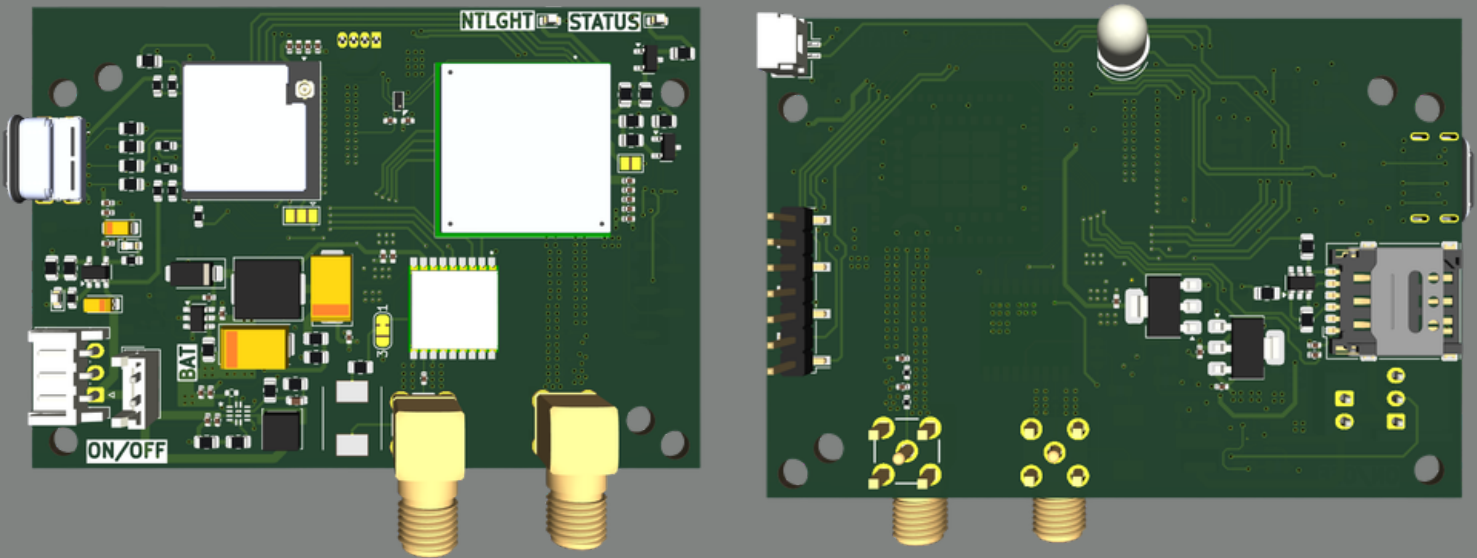
// Description:

- Designed an ultra-compact **2-layer PCB** at just **0.4mm board thickness** in **Altium Designers** for a **wearable/handheld** digital tasbeeh (prayer counter) device
- Powered by the **ATtiny1616 (low-power AVR MCU)** for efficient counting logic and **LED/motor control** in a minimal footprint
- Engineered for **3-month battery life** on a single coin-cell battery, driven by ultra-low-power firmware and efficient power conversion
- Implemented a **TPS63900 buck-boost converter** for stable voltage regulation across the full battery discharge curve – critical for long runtime on a tiny cell
- Designed battery voltage measurement circuitry for **real-time state-of-charge sensing** and low-battery detection
- Integrated **haptic feedback** via a vibration motor driver circuit for silent, tactile counting confirmation
- Designed a **4-zone capacitive/tactile LED indicator** array (dual-color LEDs) for visual count tracking and status feedback
- Added a **low-profile tactile push-button** interface for user input in a minimal-footprint, single-button UX
- Included a dedicated **UPDI (Unified Program and Debug Interface)** programming header for firmware flashing and in-system debugging
- Achieved a highly miniaturized, **single-cell-powered design** suitable for **pocket-sized** or **wrist-worn daily use**, balancing low-power embedded design with compact mechanical constraints



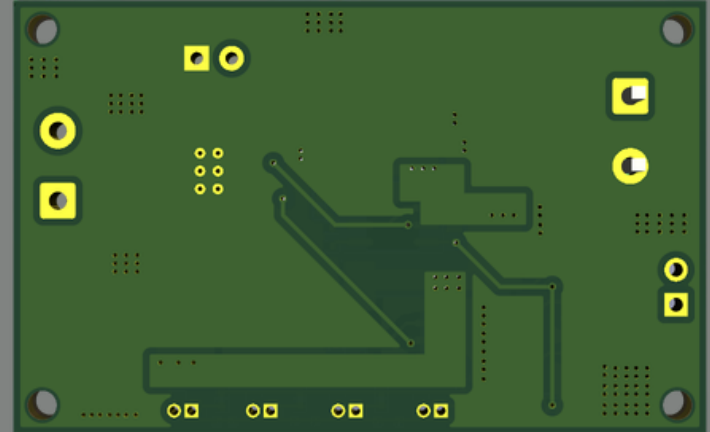
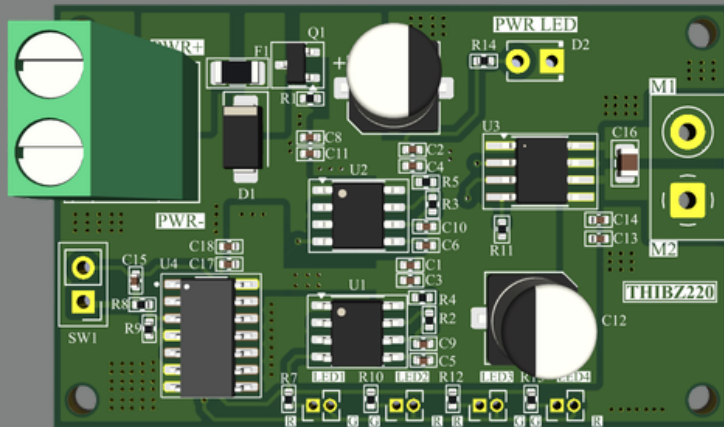
// Description:

- Designed a compact **4-layer PCB in Altium Designer** for a wearable/handheld **AI-powered text recognition device** with integrated **camera-based OCR capability**
- Built around the **ESP32-S3 (dual-core Xtensa MCU with AI vector instructions)** for on-device image capture, processing, and edge AI inference
- Integrated a camera serial interface (CSI/DVP) connector for an **OV2640-class camera module**, enabling real-time image/text capture for OCR processing
- Designed dedicated camera power rails – a 1.2V LDO and 2.8V LDO – for clean, **isolated analog/digital supply to the image sensor**
- Implemented 16MB SPI NOR flash (W25Q128) for on-device firmware and model/data storage
- Designed a complete battery power system: **Li-ion charger** (FAN54120), 3.3V LDO, and 5V boost converter (TPS61222) for peripheral/camera power
- Added a **low-power on/off power controller (LTC2950)** with single-button latch power-up/power-down sequencing
- Integrated **haptic feedback (vibration motor driver)** and a WS2812B RGB LED for tactile and visual user feedback
- Designed **RF front-end with 2.4 GHz antenna matching network for Wi-Fi/BLE wireless connectivity**, plus a logic-level shifter (SN74LV1T34) for mixed-voltage I/O
- Included a POGO-pin connector interface for compact charging/programming access, supporting a fully sealed, ultra-compact form factor



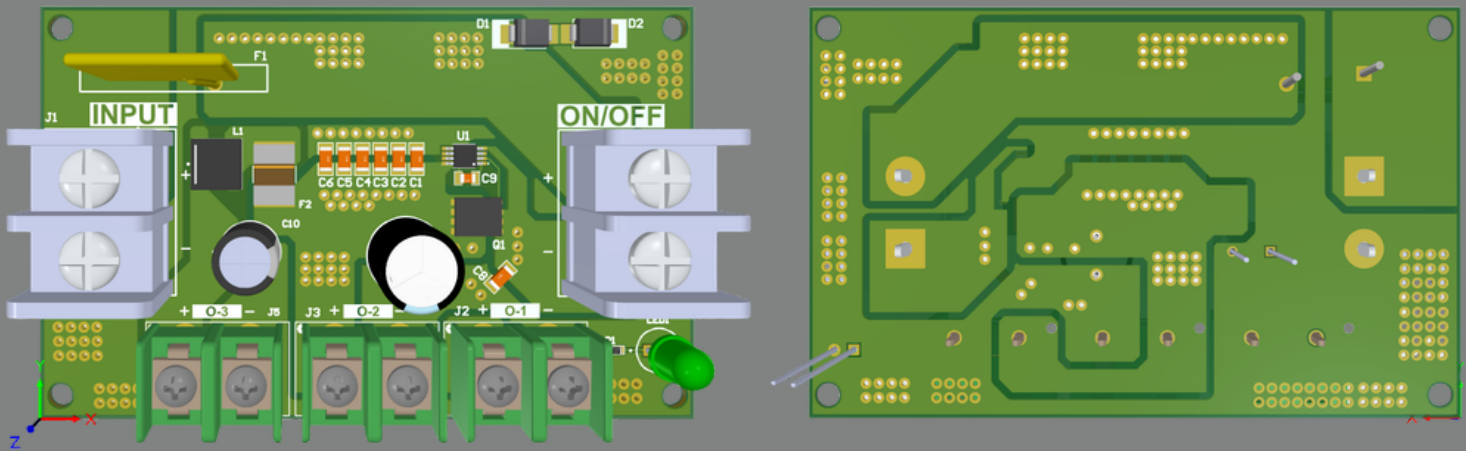
// Description:

- Designed a **4-layer PCB in KiCad 9.0** for a **cellular IoT tracking** device with integrated **LTE connectivity and GNSS positioning**
- Built around the ATmega32U4 (Atmel AVR MCU with native USB) as the host controller, managing USB communication, peripheral control, and SIMCom module interfacing
- Integrated the **A7672S SIMCom module** – a LTE Cat-1/Cat-4 cellular modem with embedded GNSS – for combined wide-area connectivity and location tracking
- Designed **microSIM card** interface with **ESD-protected** signal lines for cellular network authentication
- Implemented dual logic-level shifters (SN74LVC2T45) for safe UART/SPI voltage translation between the 3.3V MCU domain and the 1.8V modem I/O domain
- Engineered a multi-rail power architecture: MP3424 synchronous buck-boost converter, AMS1117 and AP2127 LDOs generating 1.8V/2.8V/3.3V/5V rails for RF, GNSS, and logic supplies
- Designed dedicated LTE and **GNSS RF front-ends** with SMA antenna connectors and matched RF traces (calculated microstrip impedance) for reliable **wireless link** performance
- Implemented **Li-Po battery charging** (MCP73831) with charge-status LED indication and **USB-C/connector-based** charging input
- Added status indicator LEDs (NetLight, charge status) and a WS2812B **RGB LED** for visual system/connectivity feedback
- Included **JTAG-style reset** jumpers and 16MHz crystal oscillator circuitry for reliable MCU clocking and debug/recovery access



// Description:

- Designed a compact **2-layer PCB in KiCad** for a pulsed motor/solenoid switch controller (**KATO-style turnout/switch driver**) with **bistable direction memory**
- Implemented dual **NE555 monostable timer** circuits to generate precise **50ms drive pulses**, protecting the motor/solenoid from continuous-current damage
- Used a **CD4013 dual D-type flip-flop** as an SR latch to store and toggle switch direction state (left/right) between button presses
- Integrated a **DRV8871 H-bridge motor driver** with integrated current limiting (ILIM) for safe bidirectional pulsed actuation of the switch motor
- Designed fused **12V input protection with a TVS diode** (SMAJ15A) for transient/surge suppression on the power rail
- Added **debounced mechanical switch input** for clean, reliable user-triggered state changes
- Included **multi-channel LED indicator** outputs for visual position/status feedback (left/right/active state)
- Implemented **P-channel MOSFET load switching (A03401A)** for controlled power gating
- Designed with extensive decoupling/bulk capacitance across all ICs for stable timing and clean switching transitions
- Engineered a **minimal-BOM, low-cost** control board suitable for model railway accessory automation (Kato-compatible turnout/switch machine driver)



// Description:

- Designed a **2-layer power distribution PCB in Altium Designer** for a **multi-rail 12V DC power distribution system**, supporting up to **9A on the primary output**
- Implemented **ideal diode reverse-polarity protection using the LM74610** (ideal diode controller) driving an N-channel power MOSFET (**BSC028N06NS**) for **near-zero-loss reverse-battery protection**
- **Designed dual TVS diode surge protection** (SMBJ28A/SMBJ14A) on the input rail to suppress voltage transients and protect downstream electronics
- Engineered three independently fused/distributed output rails – **12V/9A, 12V/3A, and 12V/5A** – for multi-load power distribution
- Added input fusing (blade-fuse style) for circuit-level overcurrent protection and easy field serviceability
- Designed extensive bulk and decoupling capacitance ($1000\mu\text{F}$ + multiple ceramic caps) across the main power rail for ripple suppression and load-transient stability
- Integrated an illuminated **pushbutton power switch** (EagleEye-style) for user-facing power control with visual on/off feedback
- Added a **power status LED** indicator for at-a-glance system power confirmation
- Designed with a DNP (do-not-populate) EMI filter stage (inductor + capacitor) for optional output noise filtering depending on application needs
- Used high-current **screw/wire-to-board** connectors (KF8500 series) rated for robust, vibration-tolerant power delivery – suitable for automotive, off-road, or industrial power distribution applications